THseries
Industrial Robot

MAINTENANCE MANUAL

Notice

1. Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
2. Before operating the industrial robot, read through and completely understand this manual.
3. After reading through this manual, keep it nearby for future reference.

Rev.2
April 1, 2020

TOKYO, JAPAN

Shibaura Machine

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PREFACE

This manual describes how to maintain the SHIBAURA MACHINE's TH series robot and controller. Maintenance and inspection are essential to keep the robot performance for a long time, to prevent failures and to assure safety. Be sure to look through this manual and set up a maintenance program before actually starting the robot.

This manual is composed of the following sections.

Section 1: Maintenance Items
This section describes the maintenance schedule of the robot and controller and items required for inspection and maintenance.

Section 2: Maintenance of the Robot
This section refers to the robot structure and items required for inspection and maintenance of the robot.

Section 3: Maintenance of Controller
This section refers to the robot controller structure and items required for inspection and maintenance of the controller.

CAUTIONS ON SAFETY

This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damage to assets and to assure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
! **DANGER**	This means that "incorrect handling will lead to fatalities or serious injuries."
! **CAUTION**	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)."

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.

*2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). Details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger. Details of the actual danger are indicated with pictures or words in or near the symbol.
	This means caution. The details of the actual caution are indicated with pictures or words in or near the symbol.

CAUTION

- Be sure to read through the safety instruction manual provided separately before starting actual work to assure the safe work covering from the robot installation to operation.

[Maintenance and inspection]

Be sure to observe the following items to use the robot safely.

DANGER	
Prohibited	• DO NOT incinerate, disassemble or charge the batteries. Otherwise, they may rupture.
Mandatory	• Be sure to turn off the main power switch of the controller before starting inspection or maintenance. • Batteries should be disposed of according to the user's in-house regulations.

CAUTION	
Disassembly prohibited	• The user must NEVER replace or modify parts other than those described in the instruction manual. Otherwise, the performance may deteriorate or faults or accidents will be caused.
Mandatory	• Always use the SHIBAURA MACHINE's designated spare parts when replacing the parts. • Maintenance and inspection should be performed regularly. Otherwise, the system may malfunction or accidents will be caused.

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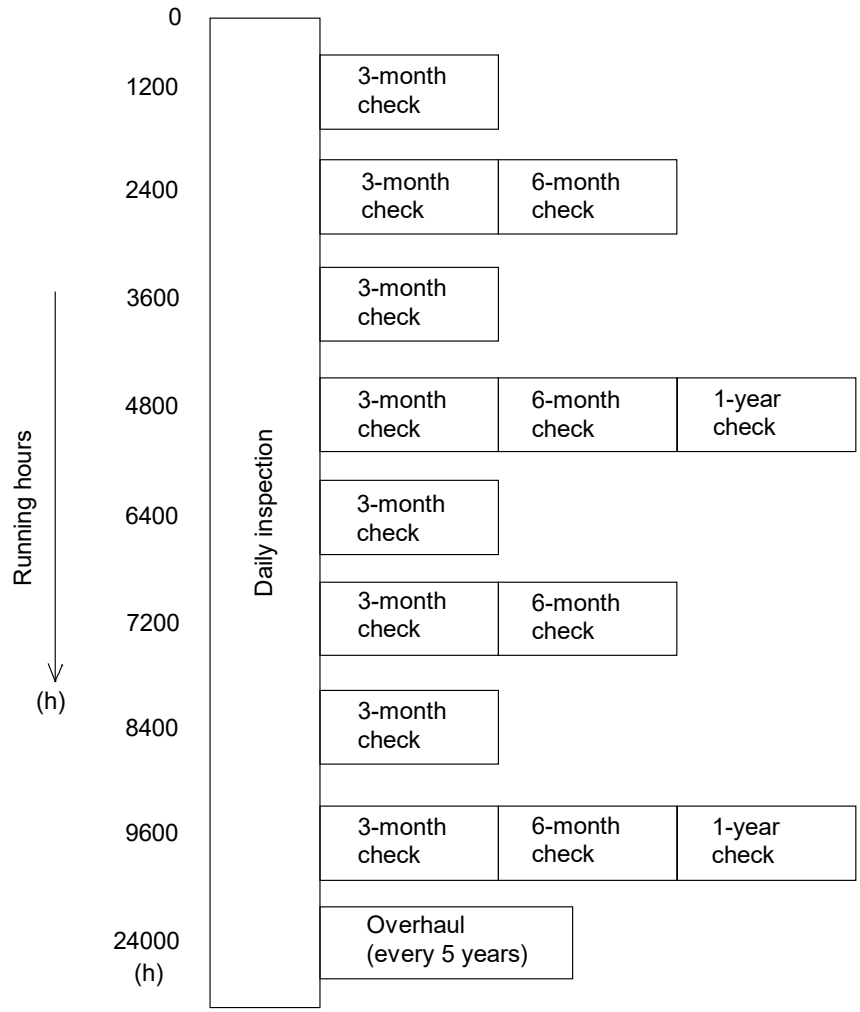
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Section 1 Maintenance Items

1.1 Maintenance Schedule

Maintenance comes in the two (2) types; daily inspection, and regular inspection and maintenance. For the regular inspection and maintenance, inspection items are added every 1,200 running hours.



Inspection schedule

- When the system is operated for 16 hours per day:
  16 hours × 25 days × 3 months = 1,200 hours
- When the system is operated for 24 hours per day:
  24 hours × 25 days × 3 months = 1,800 hours

1.2 Items for Maintenance and Inspection

This section describes the items for maintenance and inspection. For the execution procedures, see the relevant paragraph listed in the table below.

a) Daily inspection

No.	Item	Inspection procedures	Relevant Para.
1	Cable ducts	(1) Make sure that any cable duct will not twist, bend or crack locally at the time of robot motion. (2) Make sure that all connectors of the cable ducts are tightened completely.	2.3.1
2	Hand set flange	Make sure that the hand set flange bolts are tightened completely.	2.3.2
	Robot installation bolts	Make sure that the robot base installation bolts are tightened completely.	
3	Vibration and noise	Make sure that each axis can move smoothly.	
4	Change in positioning accuracy	Make sure that the robot can stop at a predetermined position accurately.	
5	Air leakage (When air piping is used)	Make sure that the air will not leak from the robot interior or air joints.	2.3.3
6	Controller vent holes	(1) Make sure that no obstacle stands in front of the vent holes. (2) Make sure that no contaminant is left in the vent hole.	3.3.1
7	Function of safety devices	Make sure that each safety device connected to the emergency stop pushbutton switch and external operation input signal can function properly.	3.3.2

b) Quarterly (every three (3)-month) inspection

No.	Item	Inspection procedures	Relevant Para.
1	Lubricated state of ball screw spline	Make sure that the ball screw spline is greased completely and that no rust is developed on it.	2.3.4

c) Semi-annual (every six (6)-month) inspection

No.	Item	Inspection procedures	Relevant Para.
1	Greasing of ball screw spline	Apply the grease to the ball screw spline.	2.3.4
2	Each axis drive system	Move each axis by hand and make sure that it moves smoothly.	2.3.5
3	Timing belts	(1) Make sure that each timing belt is not worn off or cracked. (2) Check each timing belt for tension.	2.3.6

d) Annual (every year) inspection

No.	Item	Inspection procedures	Relevant Para.
1	Greasing of ball screw spline	Apply the grease to the ball screw spline.	2.3.4
2	Greasing of reduction gears	Apply the grease from the grease nipples.	2.3.7
3	Wiring and piping	Make sure that all cables and air tubes are in the good state.	
4	Replacement of position detector battery (Replace the battery every year.)	Replace the position detector backup battery (housed in the robot base unit). If the running time is long, battery consumption is small. If the power OFF time is long, the consumption is large. If the battery voltage drops, a battery alarm occurs. Then replace the battery immediately.	2.3.8

e) Overhaul (every five (5) years)

No.	Item	Inspection procedures	Relevant Para.
1	Inspection of consumable parts	Inspect the flexible cables, reduction gears, ball screw spline, timing belts and motors and replace them when necessary.	
2	Battery replacement (Replace the battery every five (5) years.)	Replace the RAM (controller) backup battery. If the running time is long, battery consumption is small. If the power OFF time is long, the consumption is large. If the battery voltage drops, the battery alarm saying "1–145 MAIN Battery alarm" occurs. Then replace the battery immediately.	3.3.3
3	Replacement of switching power supply unit (Replace the battery every five (5) years.)	As the switching power supply unit uses an aluminum electrolytic condenser, its life is limited. Replace the switching power supply unit (DC5 V, DC24 V) for preventive maintenance. The life of the switching power supply unit differs with load current, running hours and ambient temperature. If the load current is large, the running time is long. If the ambient temperature is high, the life becomes short.	3.3.4

1.3 Maintenance Tools

We recommend to use the following tools for maintaining the robot and controller.

- Phillips screwdrivers
- Hexagonal wrench set: M3 ~ M8
- Assembly wrench set
- Water pump pliers
- Nippers
- Grease guns

1.4 Maintenance Contract and Repair

1.4.1 Maintenance Contract

With the exception of greasing and lubrication, we recommend that the user conclude with us an after-sale service contract on the inspection items for maintenance and inspection to be done semi-annually and after.

1.4.2 Repair

If a fault has occurred or if repair is necessary, turn off the controller power and contact the SHIBAURA MACHINE Service Department. At this time, advise us of the details of the fault and the following information stated on the robot and controller.

- Robot model
- Manufacture number
- Date of manufacture
- Operating time

1.4.3 Modification

This robot and controller MUST NOT be modified without the consent from SHIBAURA MACHINE.

! CAUTION

- The user must NEVER replace or modify parts other than those described in the instruction manual. Otherwise, the performance may deteriorate or faults or accidents will be caused.

Section 2 Maintenance of the Robot

2.1 Cautions on Maintenance and Inspection

When performing inspection or maintenance of the robot, strictly observe the following precautions to protect yourself and coworkers.

! DANGER

- Be sure to turn off the main power switch of the controller before approaching the robot for maintenance and inspection.

2.2 Layout of Robot Components and Drive Mechanism

The layout of the robot mechanical components is shown in Fig. 2.1.
The drive mechanism of each axis is described below.

a) Axis 1
The output of axis 1 AC servo motor is input to the harmonic drive. The output of the harmonic drive directly controls the arm 1 motion.

b) Axis 2
The output of axis 2 AC servo motor is input to the harmonic drive. The output of the harmonic drive directly controls the arm 2 motion.

c) Axis 3
The output of axis 3 AC servo motor is input to the ball screw nut through the timing belt to move the tool shaft up and down.

d) Axis 4
The revolution of axis 4 AC servo motor is input to the reduction gears. The output is input to the ball spline nut through the timing belt to revolve the tool shaft.

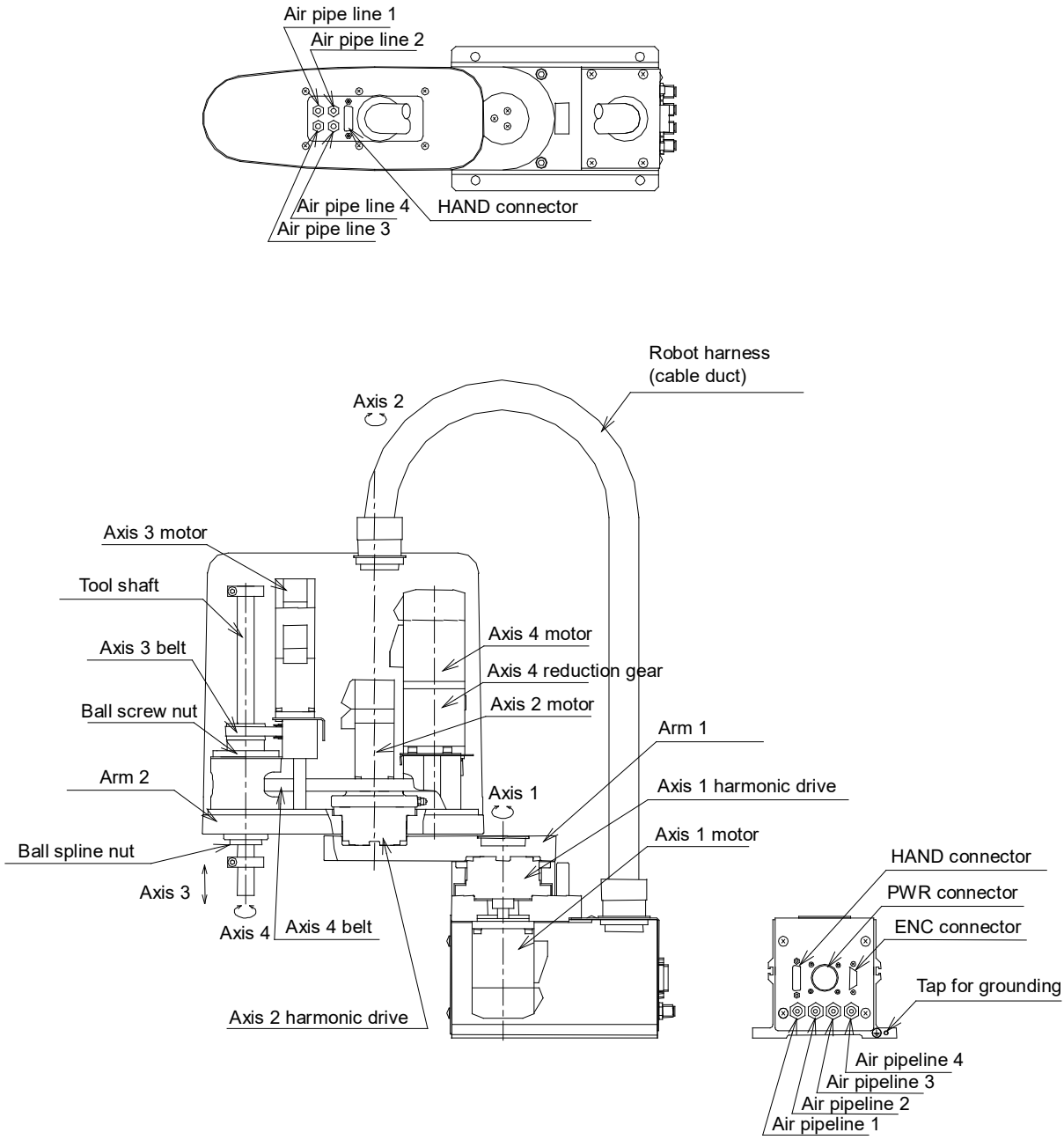


Fig. 2.1 Layout of robot mechanical components

2.3 Robot Maintenance Procedures

2.3.1 Check of Cable Duct for Abnormality

a) Visually check to see that the cable duct is not locally twisted, bent or cracked when the robot moves. If there is any abnormality, replace the cable to prevent breakage of the cable wire, etc.

b) The connectors securing the cable duct are secured by tightening it to the plate with a hexagon nut. If the hexagon nut is loosened, tighten it with a pair of water pump pliers, etc.

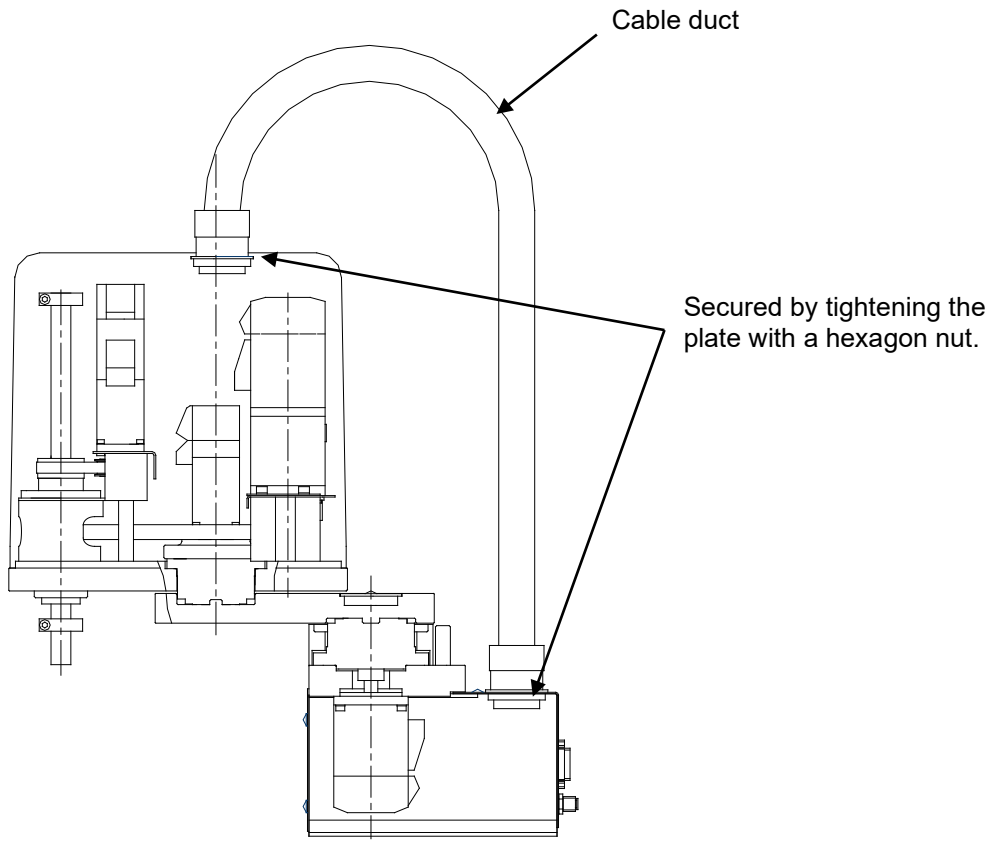


Fig. 2.2

2.3.2 Check of Hand and Base for Mounting

DANGER

- Completely tighten the hand clamp bolt. Otherwise, the hand may drop or shift from a predetermined position.

a) Check of hand set flange (option) bolt for looseness
Make sure, using a hexagonal wrench key, that the hexagon socket head cap screw of the hand set flange that is tightened onto the ball screw spline is not loosened. Otherwise, tighten it.
Also make sure by means of a hexagonal wrench key that the hexagon socket head cap screw securing the hand set flange and hand is not loosened. Otherwise, tighten it.

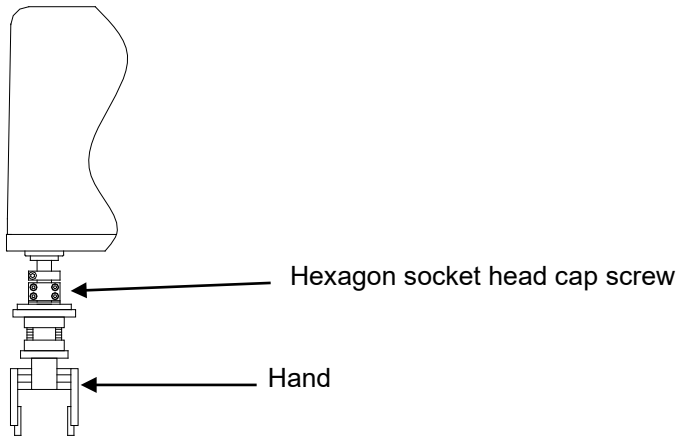
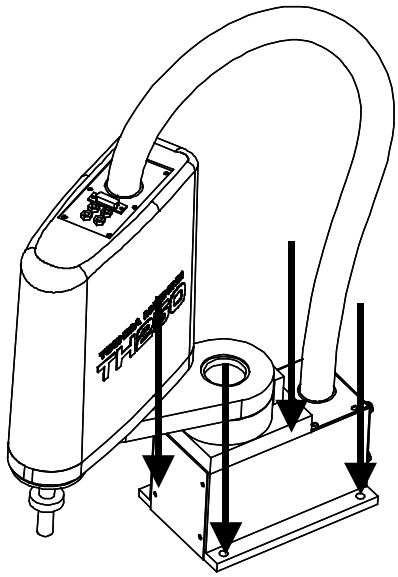


Fig. 2.3 Hand set area

b) Check of basic robot installation bolts
Make sure, using a hexagonal wrench key, that the installation bolts of the robot base are not loosened. Otherwise, tighten them.



2.3.3 Check of Air Leakage

Piping connections in the robot are shown in Fig. 2.4. Make sure that no air leaks from the robot interior or air joints.

! DANGER

- Be sure to turn off the main power switch of the controller before approaching the robot for maintenance and inspection.

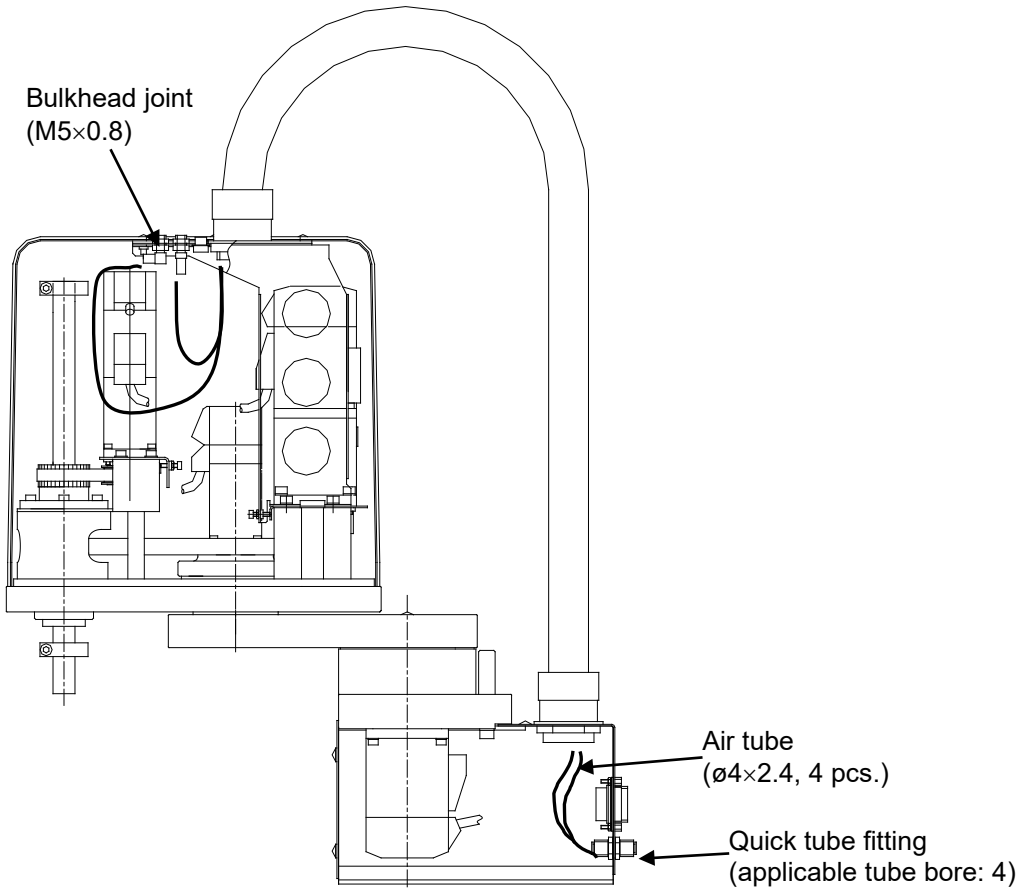


Fig. 2.4 Air piping for TH250/TH350 hand

2.3.4 Lubrication of Tool Shaft

To lubricate the tool shaft, use the following grease.

- Lithium soap based grease No.2 (Ex.: Alvania No.2 made by Shell Oil)

! CAUTION

- DO NOT mistake the type of grease. Otherwise, the robot performance may deteriorate or the robot may malfunction.
- After you have touched the ball screw spline by hand, be sure to apply the grease to it. Otherwise, rust will develop on the spline.

a) Check of ball screw spline for lubrication
Open the arm 2 cover and make sure that the tool shaft has not run out of grease and that no rust has developed on it. If the tool shaft has run out of grease, apply the grease according to Para. b).

b) Greasing of tool shaft
Apply the grease with a brush or by hand so that all guide slots around the shaft are covered. Just after the greasing, DO NOT start an operation. First, move the tool shaft slowly over the entire motion range. Then wipe off any grease that oozes from the shaft. Repeat these steps a few times.

2.3.5 Check of Each Axis Drive Status

Move each axis by hand and make sure that it can move smoothly. Perform check on the axis 3 and axis 4 while they are in the servo-free status.

! DANGER

- When nearing the robot in the servo-free status, be sure to carry out the work by two (2) persons. One person should carry out the work while the other is monitoring the work outside the hazardous zone. The latter person should watch the work and be ready to press the emergency stop switch if any abnormality occurs.

2.3.6 Check of Timing Belt Status

Remove the arm 2 cover and arm 2 lower cover, and visually check to see that the belts of axis 3 and axis 4 are not worn off or cracked.
Also check the timing belt for tension and adjust as necessary.

2.3.7 Greasing of Reduction Gears

To lubricate the reduction gears, use the following grease.

- Harmonic grease 4B No.2 (Maker: Harmonic Drive Systems)

! CAUTION

- DO NOT mistake the type of grease. Otherwise, the robot performance may deteriorate or the robot may malfunction.

Remove the axis 1 cover and axis 2 arm cover, and fill a specified volume of grease from the grease nipples, using a grease gun.
The greasing volume is 5 cm^3 and the greasing locations are shown in Fig. 2.5.
(When removing the axis 2 arm cover, see Para. 2.3.9.)
The type of grease nipple is A–M6F.
The axis 4 reduction gears are of a non-greasing type, it need not be greased at all.

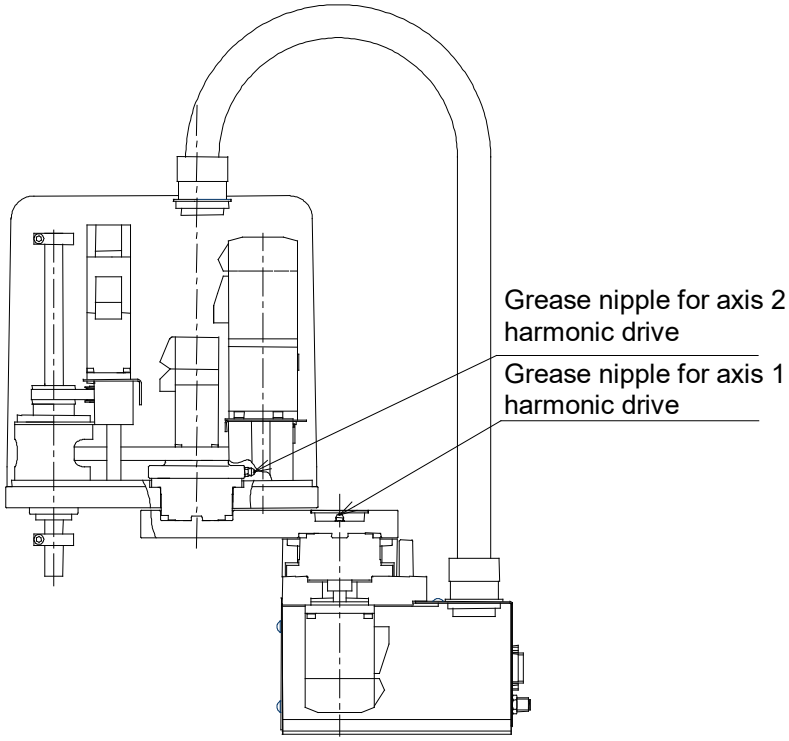


Fig. 2.5 Greasing locations of reduction gears

2.3.8 Replacement of Position Detector (or Encoder) Battery

To back up the data of the position detector attached to each motor, a battery is used. As the position detector data is backed up by the battery while the power is turned off, consumption of them varies with the running hours. If the running time per day is eight (8) hours, the life of the battery is about 2.5 years.
If the controller has detected the battery voltage drop alarm, however, replace the battery. The alarms detected by the controller at voltage drop are as follows:

1–049	Axis1 Battery alarm
1–065	Axis2 Battery alarm
1–081	Axis3 Battery alarm
1–097	Axis4 Battery alarm

The battery should be replaced while the power is turned on and emergency stop is effected after the safety is assured.

[Replacing procedures]

1) Assure the safety and press the emergency stop pushbutton switch to emergency-stop the robot while the power is turned on.
2) Remove six (6) screws M4×5 of the robot base unit to disconnect the cable connector set panel from the base unit. (Fig. 2.6a)
3) The battery is connected by means of a connector. Remove it.
4) Set a new battery.
5) Set the cable connector set panel again.
6) Reset the error from the teach pendant.

! CAUTION

- Batteries should be disposed of according to the user's in-house regulations.
- NEVER drop the batteries into fire. NEVER short-circuit, charge, disassemble or heat them. Otherwise, liquid leakage or rupture may be caused.

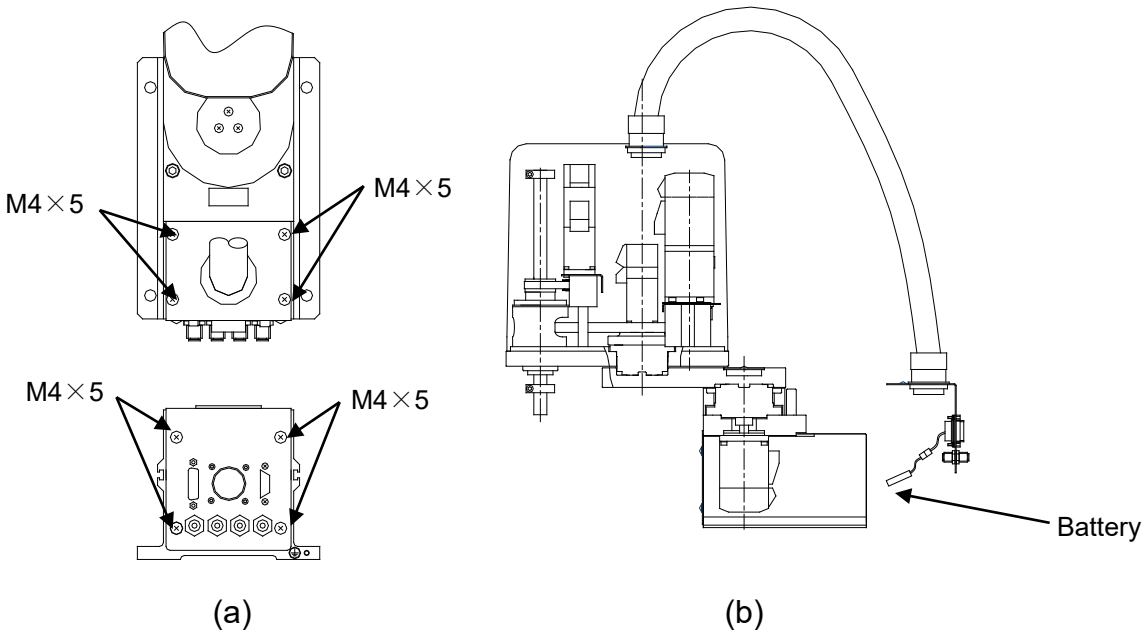


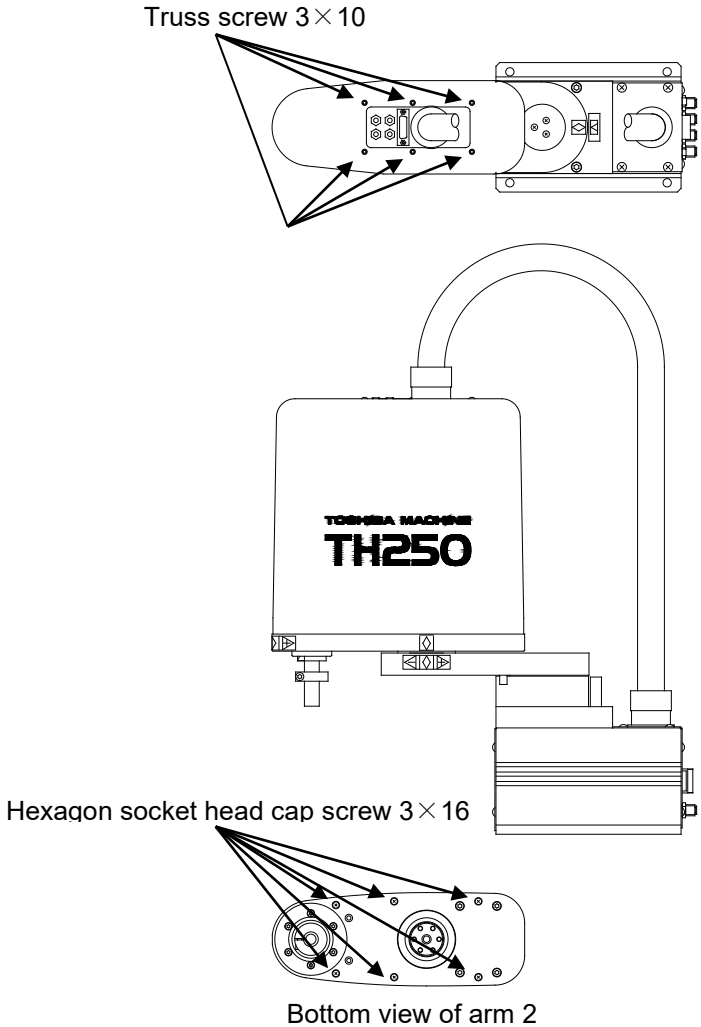
Fig. 2.6 Position detector battery set position

2.3.9 Removal of Arm 2 Cover

Provide the following tools first.

- Hexagon wrench set
- Plus screwdriver

Before starting the work, turn off the controller power.
Remove six (6) each of truss screws 3×10 and hexagon socket head cap screws 3×16. Then remove the arm 2 cover so as to draw it out in the upward direction.



Bottom view of arm 2

Section 3 Maintenance of Controller

3.1 Cautions on Maintenance and Inspection

When performing maintenance and inspection of robot controller TS1000 (hereinafter referred to as the "controller"), follow the items given below so that the work can be carried out safely.

! CAUTION

- Before removing the controller cover for maintenance or inspection, be sure to turn off the main power switch of the controller.
  Also, turn off the power breaker when connecting or disconnecting the power cable.
- DO NOT touch the servo driver for at least two (2) minutes after the main power switch is turned off. The large-capacity capacitor in the servo driver may be charged with voltage, and touching the servo driver could lead to electric shocks.
- DO NOT disconnect the battery connector at other than battery replacement. Otherwise, files saved in the memory may be lost.

! DANGER

Be sure to turn off the main power switch. If the main power switch remains ON, the servo power printed board, servo printed board and switching power supply are charged, and touching them could lead to electric shocks.

3.2 Layout of Controller Parts

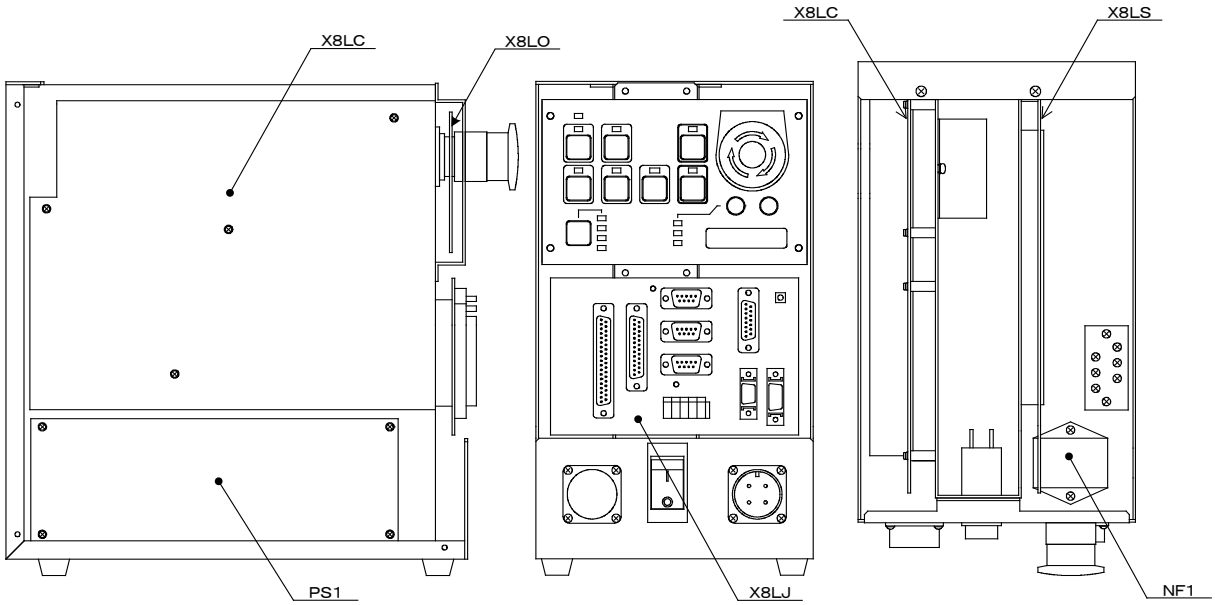


Fig. 3.1 Layout of controller parts

Part name	Descriptions
PS1	DC5 V/DC24 V switching power supply
X8LC	Main control printed board
X8LS	Servo printed board
X8LJ	Mother board concurrently serving as connector printed board.
X8LO	Control panel printed board
NF1	Noise filter

3.3 Maintenance Procedures

3.3.1 Check of Controller Air Vent Holes

If the air vent holes are blocked, the controller may overheat and malfunction. To avoid this, perform check on the air vent holes to make sure that air is flowing freely through them.

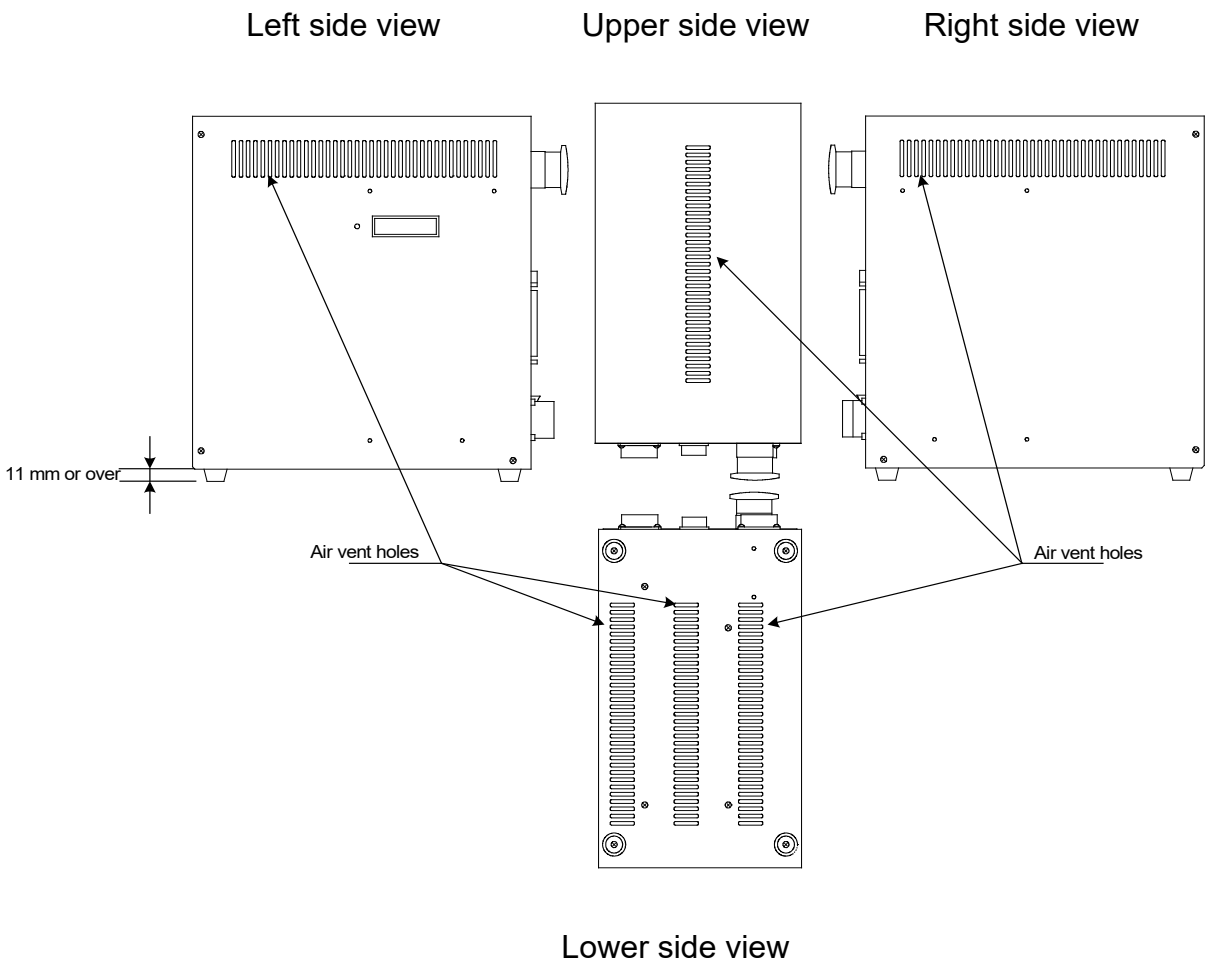


Fig. 3.2 Controller side views

a) Make sure that no obstacle stands in front of the air vent holes.
If there is any obstacles, move them away from the vent holes so that the air flow is not blocked.

b) Make sure that no contaminant is left on the air vent holes. If any contaminant is let on the vent hole, remove it.

c) If the controller is placed sideways, make sure that the air vent holes on the bottom surface of the controller are more than 11 mm away from the floor surface. Always keep the space of 11 mm or over. The length of rubber shoe is approximately 11 mm.

3.3.2 Check of Safety Devices for Function

Make sure that the emergency stop pushbutton switches equipped on the control panel and teach pendant TP1000 (hereinafter called the "teach pendant") work properly. Also make sure that the safety devices controlled by the external operation input signals work correctly.

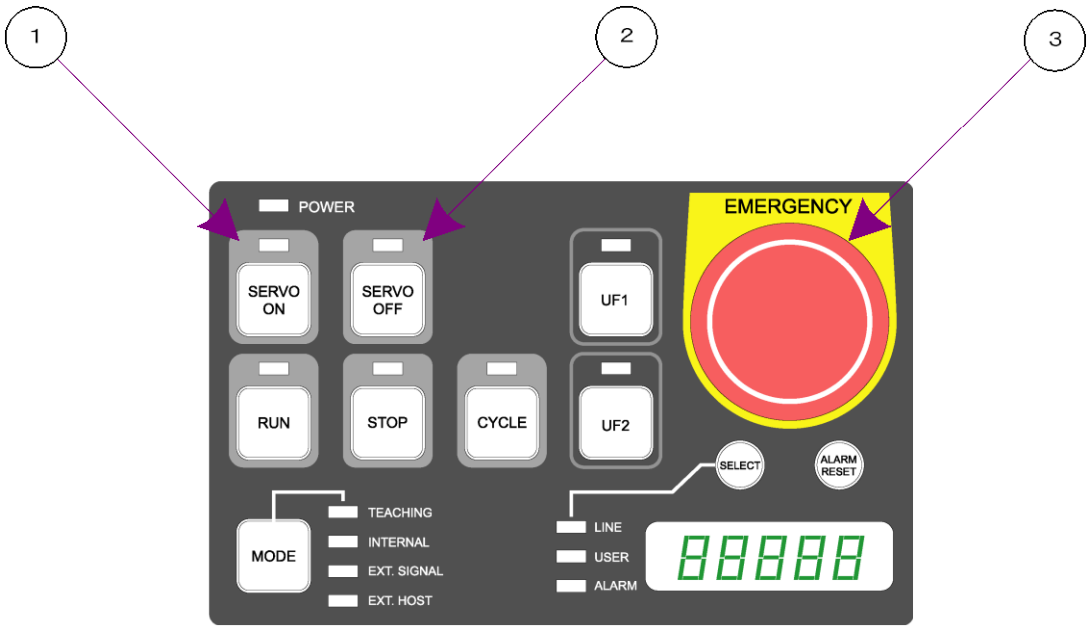


Fig. 3.3 Layout of control panel switches

a) Make sure that the EMERGENCY (emergency stop) switch [3] functions properly.

1) Press the SERVO ON switch [1] to turn the servos on and make sure that the SERVO ON switch [1] LED is illuminated.

2) Press the EMERGENCY switch [3] and make sure that the servos are turned off. The SERVO ON switch [1] LED goes out and the SERVO OFF switch [2] LED is illuminated. At this time, make sure that the EMERGENCY switch [3] remains depressed.

3) Turn the EMERGENCY switch [3] clockwise and make sure that the same switch turns off.

b) Make sure that the EMERGENCY STOP switch [5] provided on the teach pendant functions properly.

1) Press the SERVO ON switch [4] on the teach pendant to turn on the servo power, and make sure that the SERVO ON switch [4] LED is illuminated.

Note: The servo power can be turned on only in the TEACHING mode.

2) Press the EMERGENCY STOP switch [5] on the teach pendant and make sure that the servos are turned off. The SERVO ON switch [4] LED goes out.



At this time, make sure that the EMERGENCY STOP switch [5] remains depressed.

3) Turn the EMERGENCY STOP switch [5] clockwise and make sure that the same switch turns off.

c) Make sure that the safety devices controlled by the external operation signals work properly.

1) Press the SERVO ON switch [1] equipped on the control panel to turn on the servo power. Then turn on (or trip) the safety devices connected with external operation input signal "emergency stop" (such as external emergency stop switch, photoelectric type sensing safety device and foot switch) to make sure that the servo power is turned off.

2) Set ON the safety devices connected with external operation input signal "low speed command", and make sure that each signal functions properly.

3.3.3 Battery Replacement

SRAM equipped on the X8CL printed board of the controller is backed up by a lithium battery to save the data. Replace the battery every five (5) years. The lithium battery will turn its life when used for a predetermined time. If it is used, neglecting the life, the battery voltage will drop to below the voltage required for keeping the SRAM data, resulting in data being lost and faults caused by leakage of the battery liquid.

If the following alarm has occurred, replace the battery with a new one.

- 1–145 MM1 battery alarm

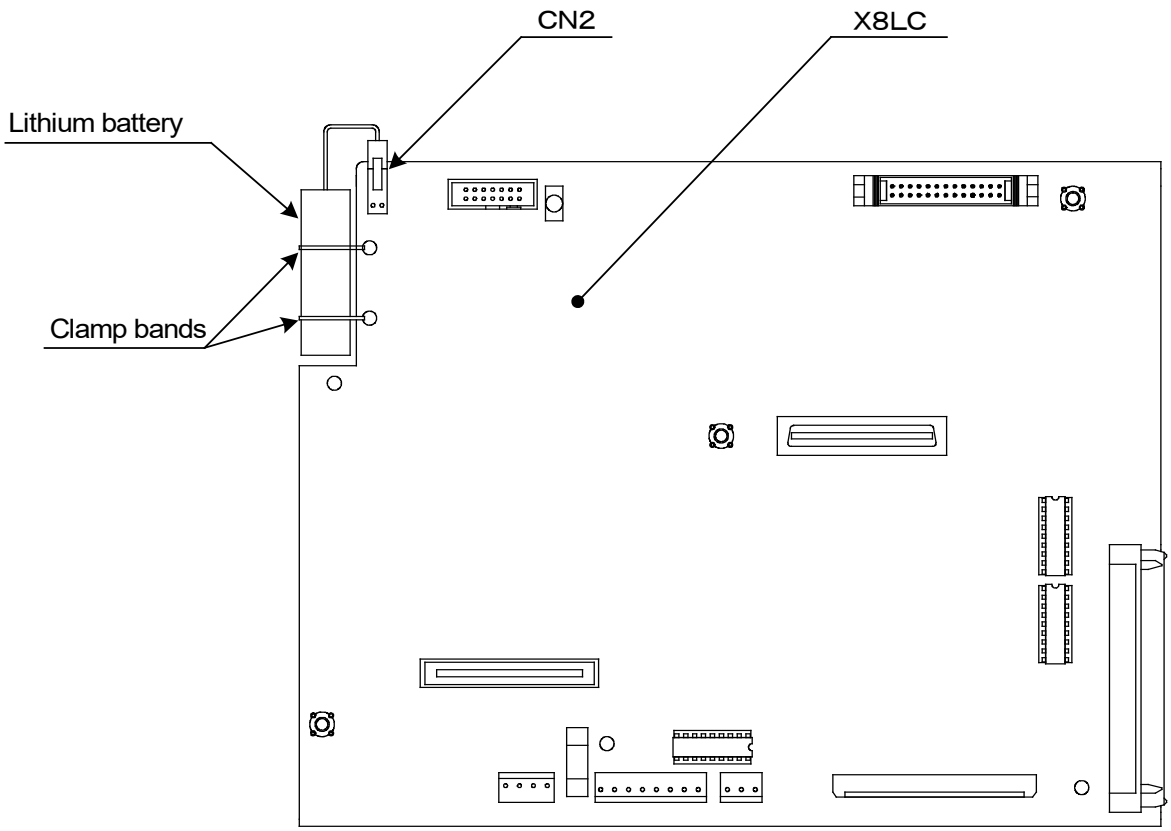


Fig. 3.5 Location of lithium battery

Table 3.1 Type of battery

Type of battery	Manufacturer	Battery service life
ER6V–C	Toshiba Denchi	10 years (at room temperature)

Note: The battery life is ten (10) years at the room temperature. As it differs with the external environment such as operating temperature and humidity, we recommend to replace the battery every five (5) years.

Only the battery shown in Table 3.1 should be used. As this is an exclusive battery, contact SHIBAURA MACHINE at order entry.

[Battery replacement procedures]

1) If the battery is kept removed for more than five (5) minutes, save all programs and parameter data stored in the internal memory in the personal computer, using the HOST connector. In the worst case, all SRAM data are lost.

2) Set OFF the MAIN POWER switch.

3) Remove the cover from the controller.

4) Remove the battery connector linked to the X8LC printed board from CN2. (For the X8LC printed board position, see Fig. 3.1.)

5) Remove the battery from the bands securing the battery. (For the battery position, see Fig. 3.5.)

6) Secure a new battery with the bands within five (5) minutes after removing the connector, then connect it with CN2.
The lithium battery capacity decreases by natural discharge even if not used. Use a lithium battery which is within three (3) years (kept at the room temperature) from purchasing.

7) Set the cover to the controller.

8) If the battery is kept removed for more than five (5) minutes, set ON the main power switch and load the programs and parameter data from the personal computer.

If the controller has detected the battery voltage drop alarm even before the five (5)-year replacement period, replace the battery immediately.

! CAUTION

- Batteries should be disposed of according to the user's in-house regulations.
- NEVER drop the batteries into fire. NEVER short-circuit, charge, disassemble or heat them. Otherwise, liquid leakage or rupture may be caused.

3.3.4 Replacement of Switching Power Supply Unit

The life of the switching power supply unit (DC5 V/DC24 V) used in the controller differs with the operating conditions. This unit uses an aluminum electrolytic capacitor, and if the load current is large and the ambient temperature is high, the life will reduce.
As the life varies largely with the user's operating conditions (5 ~ 10 years), replace the power supply unit every five (5) years for preventive maintenance.

[Replacement procedures]

1) Turn off the main power switch.

2) Turn off the power breaker.

3) Remove the cover from the controller.

4) Remove all cables from CN1 and CN2, which run from the power module to the X8LC and X8LS printed boards. (For the position of each printed board, see Fig. 3.1.)

5) Remove the power module (LEB100F–0524, made by Cosel) from the controller. (A total of four (4) clamp screws are provided.)

6) Set a new power module and clamp the four (4) screws.

7) Connect the cables with CN1 and CN2 as originally set.

8) Attach the cover to the controller.

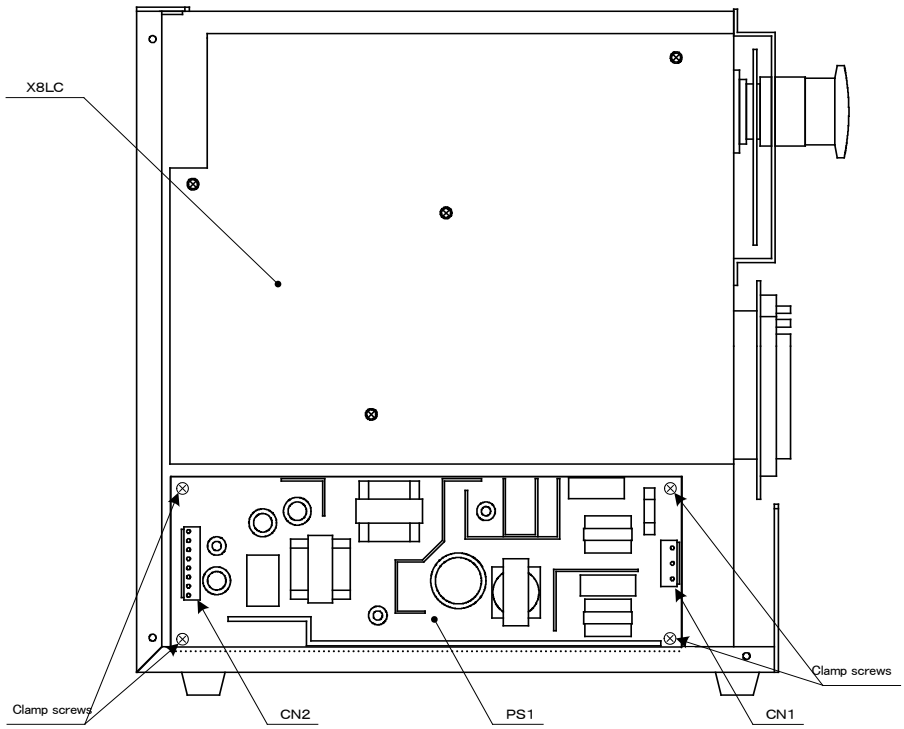


Fig. 3.6

3.3.5 Replacement of Fuse (X8LC Printed Board)

If the current exceeding the specified current has run through the I/O unit, the fuse of the X8LC printed board is blown out. If the alarm saying "X8LC fuse out (or P24V power error)" is displayed, replace the fuse with a new one.
If the fuse of the output module has been blown out, examine and identify an fault circuit, remedy the cause, then replace the fuse.

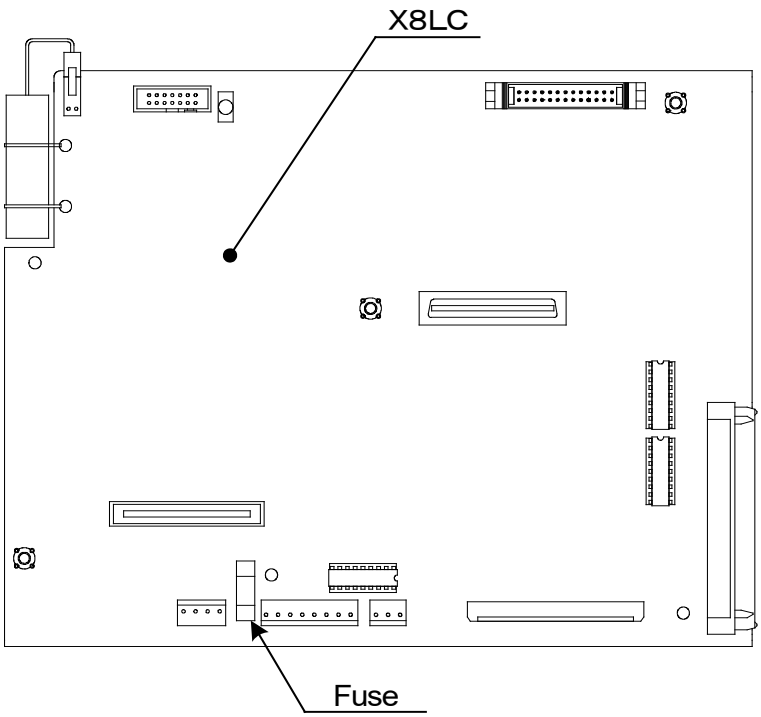


Table 3.1 Type of fuse

Type of fuse	Manufacturer
OMF–51NR 250 V 3A	World

[Replacement procedures]

1) Turn off the controller main power.
2) Remove the cover attached to the fuse unit as shown above.
3) Remove the fuse and set a new fuse.
4) Mount the removed cover.
5) Turn on the controller main power and make sure that no error will occur.

3.3.6 Replacement of Output ICs (X8LC Printed Board)

If the current exceeding the specified current has run through the output unit, the output unit may not turn on normally. (The fuse is blown out.)
When this happens, the ICs of the X8LC printed board have possibly been damaged. Replace the ICs.
Before the replacement, examine and identify an fault circuit, remedy the cause, then replace the ICs.

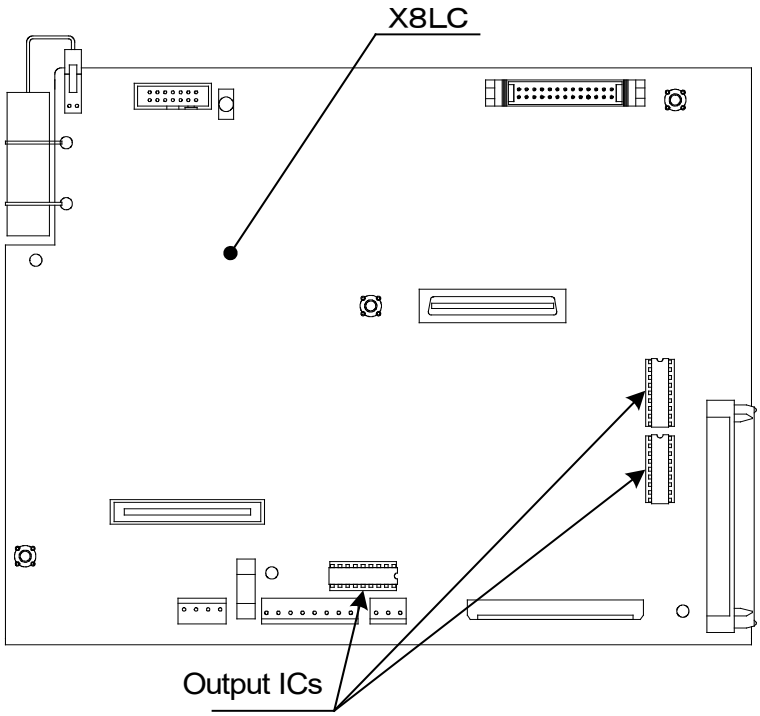


Table 3.2 Type of IC

Type of IC	Manufacturer
M54562P	Mitsubishi Electric

[Replacement procedures]

1) Turn off the controller main power.
2) Remove the ICs from respective sockets as shown above.
3) Set new ICs to the sockets.
4) Turn on the controller main power.

Attachment 1 List of Mechanical Replacement Parts for Maintenance

List of TH250 [1]/TH350 [2] replacement parts for maintenance

Part name	Model	Type	Maker	Q'ty	Remarks
AC servo motor	[1], [2]	MSMA022C2U1	SHIBAURA MACHINE	1	Axis 1
	[1], [2]	MSMA012C2U1		1	Axis 2
	[1], [2]	MSMA012C2R1		1	Axis 3
	[1], [2]	MQMA012C2D1		1	Axis 4
Harmonic drive	[1]	M074824		1	Axis 1
	[2]	M086369		1	Axis 1
	[1], [2]	M074825		1	Axis 2
Lithium battery	[1], [2]	ER6V–C		1	
Planetary reduction gears	[1], [2]	M074826		1	Axis 4
Timing belt	[1], [2]	216–3GT–9		1	Axis 3
	[1], [2]	510–3GT–12		1	Axis 4
Timing pulley	[1], [2]	S700133		1	Axis 3
	[1], [2]	S700132		1	Axis 3
	[1], [2]	S700279		1	Axis 4
	[1], [2]	S700131		1	Axis 4
Ball screw spline	[1], [2]	BNS1616AEF+300LC5		1	Axis 3
Harness	[1]	E99176G01		1	
	[2]	H02972G01		1	
Grease	[1], [2]	Harmonic grease 4B No.2	Harmonic Drive Systems	500 g 2.5 kg	
	[1], [2]	Shell Alvania grease No.2	Shell Oil	2.5 kg	

CAUTION

- With the exception of the grease, all parts listed above are custom-made to SHIBAURA MACHINE specifications. Contact SHIBAURA MACHINE at order entry.
- NEVER replace the parts other than the lithium battery by the user.

Attachment 2 List of Controller Replacement Parts for Maintenance

Part name	Type	Maker	Q'ty	Remarks
PS1 (Switching power supply)	LEB100F–0524	Cosel	1	
Main power switch	BAM205031	Matsushita Denshi	1	
Lithium battery	ER6V–C	Toshiba Denchi	1	
Fuse (X8LC printed board)	0MF–51NR 250 V 3A	World	1	
Printed circuit board	X8LC (main control)	SHIBAURA MACHINE	1	
	X8LS (servo)		1	
	X8LJ (mother board concurrently serving as connector board)		1	
	X8LO (control panel)		1	
Teach pendant	TP1000	SHIBAURA MACHINE	1	With 5 m-long cable
IC	M54562P	Mitsubishi Electric	3	

Appendix Diagram 1 Component Connection Diagram

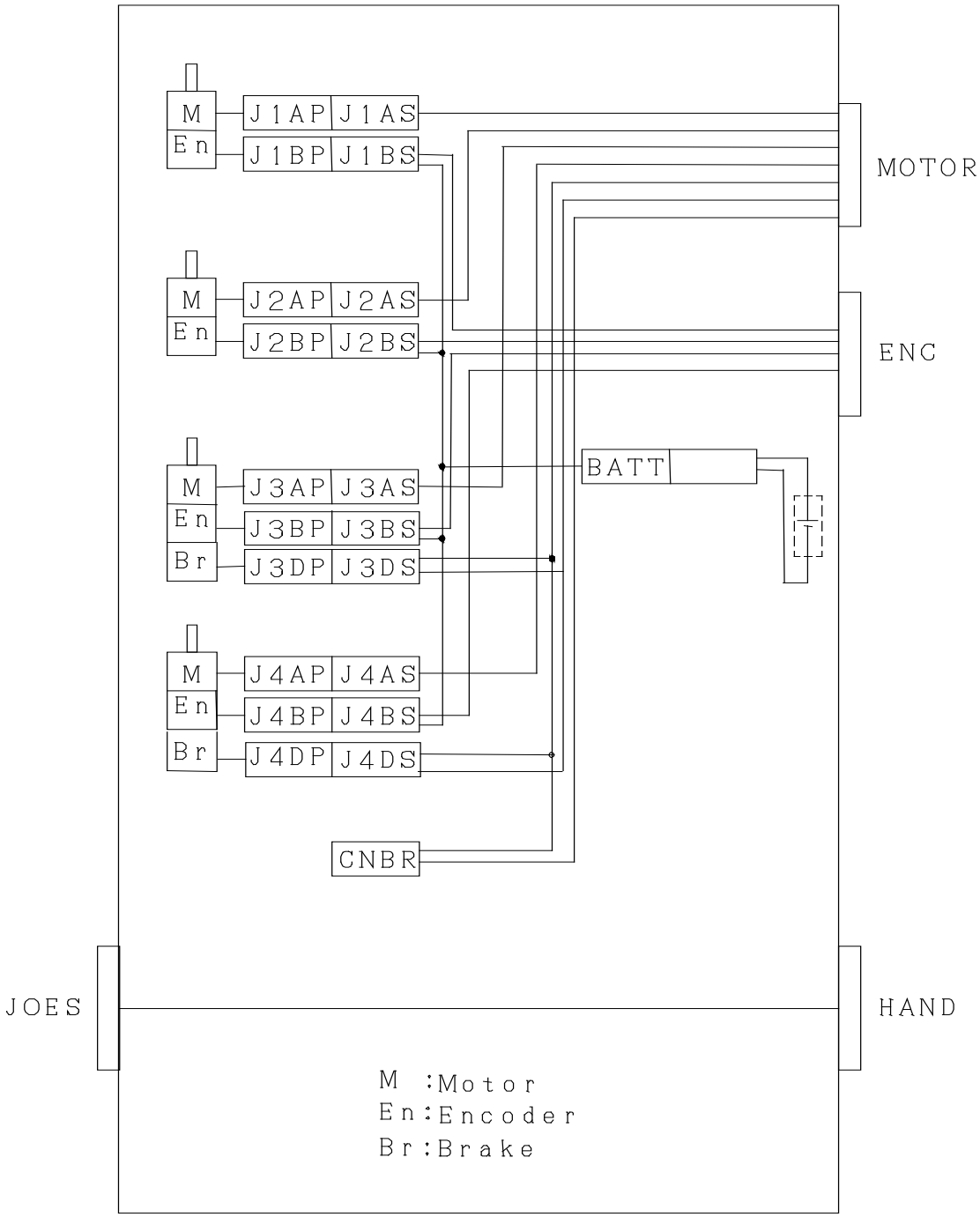


Fig. 4.1 Component connection diagram

Appendix A Restoring Home Position Data

Batteries which back up the positional information of the position detectors (i.e., encoders) are contained in the robot base unit. Unless the robot is used for a long period of time, the battery voltage drops and the home position data will be lost. Also, if the voltage of the battery which backs up the internal memory of the X8LC printed board drops, programs, parameters and absolute coordinate data set by the user will be lost. To avoid this, replace the battery on a regular basis. The yardstick of each replacement period is shown below.

- Batteries for position detectors ··· Two (2) and a half (2.5) years after delivery (when the operating time per day is eight (8) hours).
- Battery for X8LC printed board SRAM ··· Five (5) years after delivery

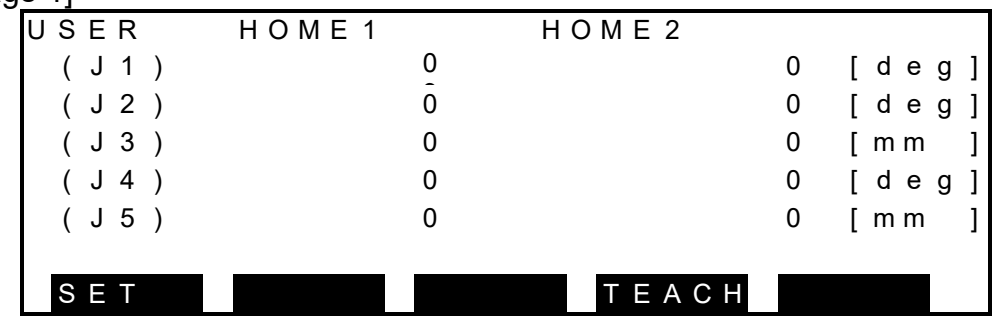
If the home position data has been destroyed due to battery voltage drop, etc., observe the following procedures to restore it.

a) How to restore the home positions

(1) Descriptions of HOME function

This function is useful to restore the original home position data by memorizing each axis coordinate data beforehand, if the machine home position data has been destroyed.
The set points can be assigned to HOME1 ~ 4. The plus (+) side mechanical stopper positions of axes 1 ~ 3 are factory-set in HOME3, and the minus (–) side mechanical stopper positions of axes 1 ~ 3 are factory-set in HOME4. The axis 4 is set at the 0° position according to the match-mark. In HOME1 and HOME2, the user can specify any desired positions. We recommend to set there the positions securing the robot end to prepare for the system construction.

[Page 1]



[Page 2]

```
SYSTEM   HOME3          HOME4
  (J1)    1183468   -    1180474   [deg]
  (J2)    1450716   -    1450720   [deg]
  (J3)    1239092   -      36711   [mm ]
  (J4)          0              0   [deg]
  (J5)          0              0   [mm ]

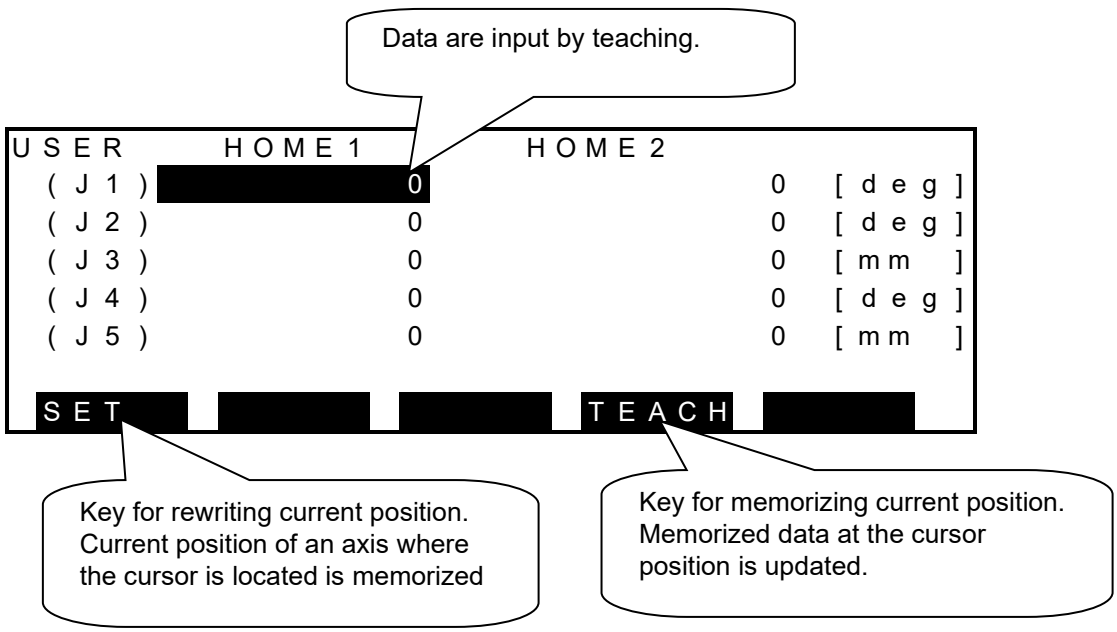
 SET                              TEACH
```

Note: The above are the screens for restoring the machine home positions. NEVER operate them at other than the restoration.

(2) Operating procedures

I. How to set the data in HOME1 and HOME2:
This paragraph describes the data setting in HOME1. (The same procedures are applied to HOME2.)

[1] Guide the robot to secure the tool shaft at a desired position.
[2] Turn off the servo power.
[3] Press the [UTILITY] key through the teach pendant.
[4] Press the NEXT key until [REORG] appears, then press the [REORG] key.



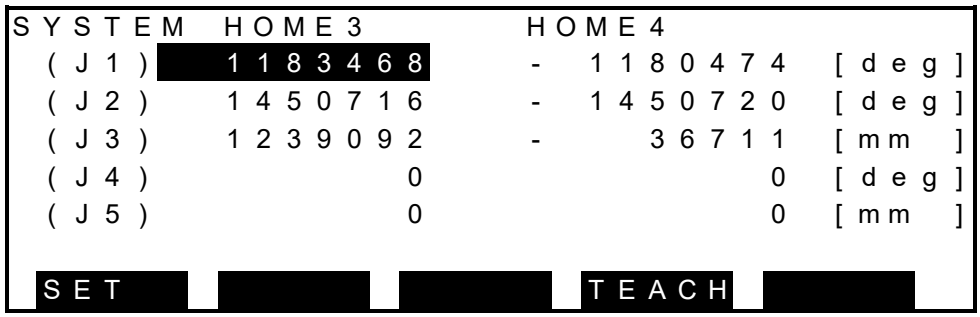
[5] When the above screen appears, memorize the position where the robot was secured in the [TEACH] mode, by moving the cursor to each axis.

II. How to restore the home position data by HOME1 (or HOME2):
If the machine home position coordinate data has been destroyed, the machine home position can be restored by means of the HOME1 function.

[1] Guide the robot to the position set by HOME1 and secure the robot there.
[2] Turn off the servo power.
[3] Press the [UTILITY] key through the teach pendant.
[4] Press the NEXT key until [REORG] appears, then press the [REORG] key.
[5] Move the cursor to each axis in the HOME1 mode of the REORG screen. Press the [SET] key and write the position where the robot was secured to the coordinate value set in HOME1. Thus, the machine home position can be restored. When you are going to set the data to axis 3 at this time, be sure to set axis 4 first.

III. How to restore the position data by HOME3 (or HOME4):
The plus (+) side mechanical stopper positions of axes 1 ~ 3 are factory-set in HOME3, and the minus (–) side mechanical stopper positions of axes 1 ~ 3 are factory-set in HOME4. The axis 4 is set at the 0° position according to the match-mark.
The machine home position can be restored, using these data.
This paragraph deals with the restoration procedures, using HOME3.

[1] Turn off the servo power.
[2] Press the [UTILITY] key through the teach pendant.
[3] Press the NEXT key until [REORG] appears, then press the [REORG] key.
[4] Press the NEXT key to display Page 2.



[5] Contact the axis 1 of the robot to the plus (+) side mechanical stopper by hand.

[6] Move the cursor to (J1) of HOME3, then press the [SET] key. Now, the machine home point of axis 1 is restored.

[7] Likewise, contact the axis 2 of the robot to the plus (+) side mechanical stopper by hand.

[8] Move the cursor to (J2) of HOME3, then press the [SET] key. Now, the machine home point of axis 2 is restored.

[9] Press the [ESC] key to escape from the REORG screen once.

[10] Turn on the servo power, and set the axes 3 and 4 in the servo-free state.

[11] Set the axis 4 to the match-mark.

[12] Contact the axis 3 of the robot to the plus (+) side mechanical stopper by hand.

[13] Turn off the servo power.

[14] Repeat Steps [2] to [4] inclusive to call the HOME screen again.

[15] Move the cursor to (J4) of HOME3, then press the [SET] key. Now, the machine home point of axis 4 is restored.

[16] Move the cursor to (J3) of HOME3, then press the [SET] key. Now, the machine home point of axis 3 is restored.

[17] Press the [ESC] key to escape from the HOME screen. Now, the operation has finished.